AYUSH KASHYAP

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Summary

AI and backend engineer who has shipped production **LLM systems** — cutting **API latency from 1.8s to under 600ms** and improving **RAG retrieval precision by 28%** at a construction-tech startup. Active open source contributor to **Ray** and **Milvus**, with 4+ merged PRs. Comfortable owning systems spanning **ETL pipelines**, **multi-agent orchestration**, **WASM-compiled engines**, and **reinforcement learning agents**.

Experience

Software Engineer · Constructure AI — Dec 2025 – Apr 2026 · Remote

- Architected **RESTful microservices** for LLM-driven construction software, reducing **p95 API latency from 1.8s to under 600ms** via Redis caching, connection pooling, and async job queues.
- Built scalable **ETL pipelines** processing **500MB+ BIM/IFC datasets** for downstream LLM context generation and structured retrieval.
- Developed **multi-agent orchestration** with **OpenAI SDK**, coordinating **4+ specialized agents** while reducing cross-agent error propagation by **40%**.
- Implemented **hybrid RAG retrieval** combining **pgvector semantic search** with structured SQL filtering, improving retrieval precision by **28%**.

Open Source Contributions

Ray (ray-project/ray) — *4+ merged PRs* · *PR #63695* · *PR #63952*

- Fixed **AlgorithmConfig.to_dict()** for RLlib's New API Stack — stale `train_batch_size` values caused incorrect experiment logging across Ray Tune trials; resolved a read-only property conflict preventing safe `update_from_dict()` round-trips, adding regression tests covering the DreamerV3 edge case.
- Patched a **critical cluster resilience bug** in Ray Core's C++ OOM killer: under memory pressure, `TimeBasedWorkerKillingPolicy` killed the `ServeController` (oldest actor on node), triggering an infinite OOM death-loop; introduced a generic `is_system_actor` shield plumbed from **protobuf** → **C++ raylet** → **Cython** → **Python**, with cross-boundary unit tests.

Milvus (milvus-io/milvus) — PR #50666 · Open · *3+ PRs; ongoing*

- Extended the **vector database filter expression engine** to support bitwise AND (&), OR (||), and XOR (ˆ) operators for bitmask queries; modified the full pipeline — protobuf schema, ANTLR parser, C++ bitset engine, and JSON/array/index execution paths (**500+ lines across 8 files**).

Projects

GoChess Engine — Demo

- Built a fully playable chess engine in **Go**, implementing **Minimax Search** with **Alpha-Beta Pruning** that reduces the effective game-tree by **60%** and completes depth-5 analysis in **under 200ms**; cross-compiled to **WebAssembly (WASM)** and offloaded search to **Web Workers**, eliminating UI jank and achieving **2–3×** throughput over a single-threaded baseline.
- Integrated **Zobrist Hashing** with a **Transposition Table** caching **100K+ positions/session** to avoid re-evaluating identical board states, and added **Quiescence Search** to prevent the horizon effect and improve tactical accuracy in volatile positions.

PyGit — Distributed Version Control Engine — GitHub

- Re-implemented core **Git internals** (Blob, Tree, Commit) with **content-addressable storage** and **SHA-1 persistence**; supported repositories with **100+ commits** while maintaining **O(1)** object lookup.
- Built a CLI supporting **init, add, commit, log**, demonstrating understanding of **DAG-based version tracking**.

PacmanRL — Deep Reinforcement Learning Agent — GitHub

- Trained a **DQN agent** with experience replay achieving stable gameplay within **500 episodes**; custom **reward shaping** reduced episode length by **30%** and delivered **2×** higher scores over random baselines.
- Engineered an **11-dimensional game-state feature pipeline** enabling policy generalization across unseen maze layouts.

Technical Skills

Languages: Go, TypeScript, JavaScript, Python, C++, C#, Java

AI / ML: LangChain, LangGraph, RAG, pgvector, Pinecone, Milvus, TensorFlow, DQN, OpenCV, Multi-Agent Systems, Prompt Engineering

Backend & APIs: Node.js, Express.js, REST APIs, GraphQL, Microservices, Redis, PostgreSQL, ETL Pipelines, Async Processing

Frontend & Web: React, Next.js, React Native, WebAssembly (WASM), Web Workers

DevOps & Infra: Git, Docker, CI/CD, Distributed Systems, Ray, Performance Profiling

Education

Punjab Engineering College (PEC), Chandigarh — 2023 – 2027

B.Tech in Mechanical Engineering — CGPA: 8.28 / 10